

Chapter 9: Essential Probability Rules

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Part II: Probability and Distributions

Randomness and Probability

When people hear the word “random” they often think of something completely haphazard or patternless. In statistics, randomness has a more precise meaning.

Random phenomenon

A phenomenon is **random** when individual outcomes are uncertain but a regular and predictable distribution emerges over many repetitions.

Empirical Probability

The probability of an outcome is the proportion of times that outcome would occur in a very long series of repetitions.

Example: Radioactive Tracers

A medical scan uses a radioactive tracer. We cannot predict which individual atom will decay next. Yet among millions of atoms, the fraction that decays over a fixed period is remarkably predictable. Probability describes this *long-run regularity*.

Two Ways to Determine Probability

How do we decide what probability to assign to an event? In practice, we usually rely on one of two approaches.

Theoretical probability

Derived from an underlying mathematical or physical model.

Example: A model of radioactive decay tells us that, during one half-life

$$P(\text{atom decays}) = 0.50$$

The probability comes from the **model**, not from counting observations.

Empirical probability

Estimated from the frequency observed in data.

Example: If 498 of 1000 tracer atoms decay during one half-life, then

$$P(\widehat{\text{atom decays}}) = \frac{498}{1000} = 0.498$$

The probability is estimated from **data**.

Both approaches can be used to construct a **probability model** and both must obey the same mathematical probability rules.

The Frequentist Interpretation

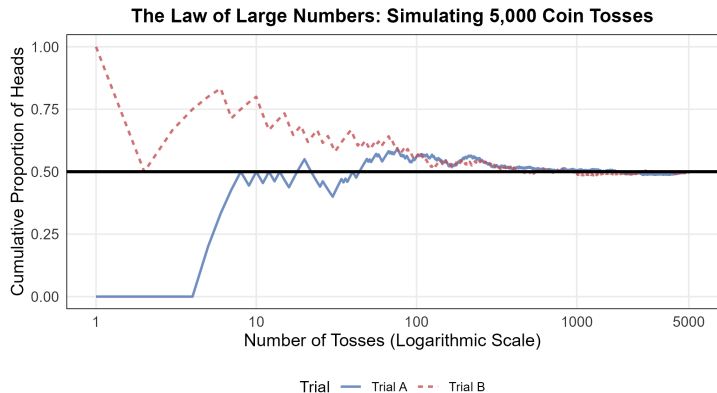
Core idea

In the **frequentist interpretation**, probability describes the proportion of times an outcome occurs when a trial is repeated many times under the same conditions. Any single outcome is uncertain, but the **relative frequency** tends to settle near its actual probability as the number of trials grows.

Radioactive tracer example

Suppose 498 of 1000 tracer atoms decay during one half-life: a proportion of 49.8%. A fresh group of 1000 atoms would probably give a slightly different result. Across many groups, the proportions would tend to cluster around the model's prediction of 50%.

Coin Toss Simulation



- The **cumulative proportion of heads** after n tosses is
$$\frac{\text{number of heads in the first } n \text{ tosses}}{n}$$
- Each line tracks this proportion through a separate sequence of 5000 tosses.
- Early proportions can swing sharply because only a few tosses have occurred.
- As the number of tosses grows, both proportions tend toward 0.5.
- The horizontal axis is logarithmic so the early variation is easier to see.

The Law of Large Numbers

Law of Large Numbers

As the number of independent repetitions increases, the observed relative frequency tends to approach the underlying probability.

- Early results may fluctuate dramatically because each new observation has a large effect.
- With thousands of repetitions, one additional result has very little influence on the cumulative proportion.
- This long-run stabilization is one reason large samples are generally more dependable than very small samples.

Important caution

The Law of Large Numbers describes **long-run behaviour**. It does not force a short sequence to “balance out”. For example, a coin has no memory of previous tosses.

The Illusion of Randomness

Which sequence is more likely when a fair coin is tossed six times?

Sequence A

H T H T T H

Sequence B

T T T H H H

The Illusion of Randomness

Which sequence is more likely when a fair coin is tossed six times?

Sequence A

H T H T T H

Sequence B

T T T H H H

Answer

Both specific sequences have probability

$$\left(\frac{1}{2}\right)^6 = \frac{1}{64}$$

True randomness naturally produces streaks and clusters.

Definition

A **probability** is a number between 0 and 1 that expresses how likely an event can happen based on the available information.

Example

What is the probability that humans discover extraterrestrial life during this century?

This event cannot be repeated thousands of times. The probability must therefore be based on evidence, scientific models and expert judgement rather than direct long run repetition.

Regardless of its interpretation, every valid probability must obey the mathematical rules introduced later in this chapter.

Vocabulary of Probability Models

To use probability mathematically, we need to describe both the possible outcomes and how likely those outcomes are.

Sample space

The **sample space** S is the set of *all* possible outcomes of a random phenomenon.

Event

An **event** is one outcome or a collection of outcomes; an event is a subset of S .

Probability model

A probability model specifies:

- 1 a sample space S
- 2 a rule for assigning probabilities to events

For example, if we are tossing a coin, the sample space is $S = \{\text{Heads}, \text{Tails}\}$.

Example 9.5: Blood Types

For the eight blood types

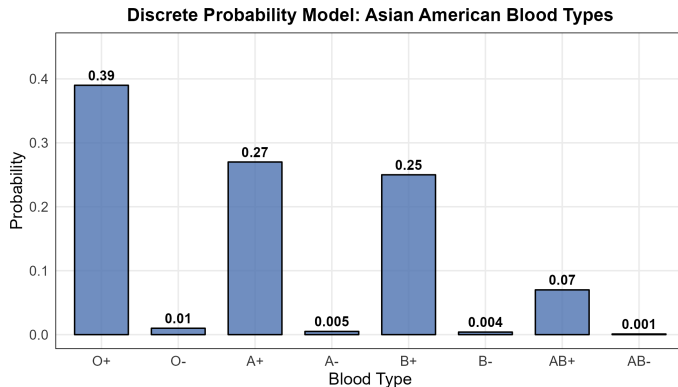
$$S = \{O+, O-, A+, A-, B+, B-, AB+, AB-\}$$

Type	$O+$	$O-$	$A+$	$A-$	$B+$	$B-$	$AB+$	$AB-$
Probability	0.39	0.01	0.27	0.005	0.25	0.004	0.07	0.001

- Each listed outcome is discrete.
- The probabilities are nonnegative and sum to 1.

$$\sum_{i=1}^{n=8} \text{probability}(\text{blood type}_i) = 0.39 + 0.01 + 0.27 + 0.005 + 0.25 + 0.004 + 0.07 + 0.001 = 1$$

Blood-Type Probability Model



The bars visualize the probability assigned to each discrete outcome.

- $O+$ has the largest probability in this model.
- Rare outcomes such as $AB-$ still belong to the sample space.
- Adding the eight probabilities gives exactly 1.

Changing the Scope of a Model

The Rhesus (Rh) factor describes whether red blood cells have a marker called the RhD antigen. If the marker is present, the blood type is $Rh+$ and if it is absent the blood type is $Rh-$. Suppose we record only the Rh factor, ignoring whether the blood type is A, B, AB, or O. Then our sample space is

$$S = \{Rh+, Rh-\}$$

Rh factor	$Rh+$	$Rh-$
Probability in a hypothetical population	0.98	0.02

Key idea

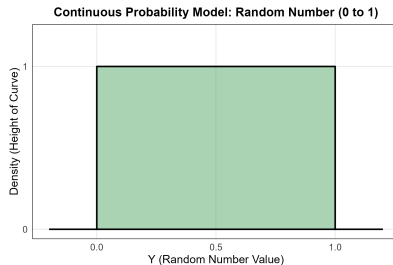
The sample space depends on how the random phenomenon is defined. A broader model can often be collapsed into a simpler model by combining outcomes.

Example 9.6: A Random Number from 0 to 1

Let Y be a random number selected from the interval $[0, 1]$.

$$S = \{y : 0 \leq y \leq 1\}$$

- There are infinitely many possible decimal values.
- We cannot list every possible value and attach a positive probability to each one.
- Instead, probabilities are assigned to **intervals** as areas under a density curve.
- The entire area under the density curve must equal 1.



Discrete vs. Continuous Sample Spaces

Discrete

- Finite or countable outcomes
- Individual outcomes may have positive probability
- Probabilities can be listed in a table
- Example: blood type

Continuous

- An unbroken interval of values
- Individual points have probability 0
- Probabilities are areas under a curve
- Example: exact height

Probability Rules

These rules must hold for **every** valid probability model, whether the sample space is discrete or continuous.

- ① **Sample space has probability 1** Given a sample space S

$$P(S) = 1$$

- ② **Probabilities are proportions:** For an event $A \in S$

$$0 \leq P(A) \leq 1$$

A probability can never be negative or exceed 1.

- ③ **Disjoint events** Two events $A \in S$ and $B \in S$ are disjoint if

$$P(A \text{ or } B) = P(A) + P(B)$$

- ④ **Complement of an event** For an event $A \in S$

$$P(A^c) = 1 - P(A)$$

Probability Rules

Two events are **disjoint** when they have no outcomes in common.

$$P(A \text{ and } B) = 0$$

Therefore

$$P(A \text{ or } B) = P(A) + P(B)$$

For three disjoint events A, B and C

$$P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C)$$

so

$$P(A \text{ and } B) = 0 \quad P(A \text{ and } C) = 0 \quad P(B \text{ and } C) = 0$$

Do not double-count

This simple addition rule applies only when the events cannot occur together.

OR

Inclusive: at least one event occurs.

A alone, B alone or both may occur.

AND

Intersection: both events occur at the same time.

This corresponds to the common outcomes of A and B .

When events overlap, the general addition rule is needed to correct for double counting. That rule is introduced in Chapter 10.

Example 9.7: Positive Rhesus Factor

The positive blood type outcomes are disjoint:

$$O+ \quad A+ \quad B+ \quad AB+$$

Therefore

$$\begin{aligned} P(Rh+) &= P(O+) + P(A+) + P(B+) + P(AB+) \\ &= 0.39 + 0.27 + 0.25 + 0.07 \\ &= \boxed{0.98} \end{aligned}$$

Example 9.7: Complement Rule

The complement of having a positive Rhesus factor is having a negative Rhesus factor.

$$\begin{aligned}P(Rh-) &= 1 - P(Rh+) \\&= 1 - 0.98 \\&= \boxed{0.02}\end{aligned}$$

Why complements are useful

It is often easier to subtract one probability from 1 than to add many separate probabilities.

Random Variables

Random phenomena often produce numerical outcomes. We use a random variable to give those outcomes a convenient mathematical name.

Definition

A **random variable** is a variable whose numerical value is determined by the outcome of a random phenomenon. We usually denote random variables by uppercase letters such as X or Y .

Probability distribution

The probability distribution of X specifies:

- which values X can take;
 - how probabilities are assigned to those values.
-
- X = number of impaired ears in a Dalmatian: $X \in \{0, 1, 2\}$.
 - Y = random number selected from $[0, 1]$: any real value in the interval is possible.

Two Types of Random Variables

Discrete random variable

Takes values from a finite or countable set.

Example:

$$X \in \{0, 1, 2\}$$

Continuous random variable

Can take any value in an interval.

Example:

$$0 \leq Y \leq 1$$

A measured variable can be intrinsically continuous even when recorded in rounded units.

Definition

A discrete probability model has a finite or countable list of outcomes. The probability of an event is the sum of the probabilities of the outcomes belonging to that event.

A valid discrete probability model must satisfy:

$$0 \leq p_i \leq 1 \quad \text{and} \quad \sum_i p_i = 1$$

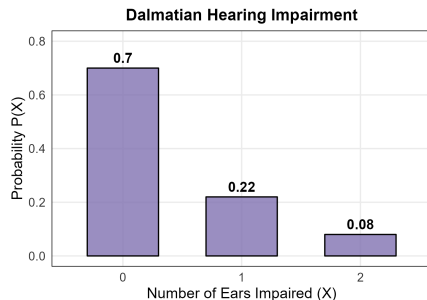
Example 9.8: Hearing Impairment in Dalmatians

A study of 5333 Dalmatians recorded the number of ears with hearing impairment. Let X be the number of impaired ears for a randomly selected dog.

X	0	1	2
$P(X = x)$	0.70	0.22	0.08

The possible outcomes are the whole numbers 0, 1, and 2. So X is a **discrete random variable**. Notice that

$$0.70 + 0.22 + 0.08 = 1$$



Example 9.8: Probability of Some Hearing Impairment

What is the probability that a randomly selected Dalmatian has hearing impairment in at least one ear?

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What is the probability that a randomly selected Dalmatian has hearing impairment in at least one ear?

For a randomly selected Dalmatian, “some hearing impairment” means **at least one** impaired ear. That is, $X = 1$ or $X = 2$.

$$\begin{aligned}P(X \geq 1) &= P(X = 1) + P(X = 2) \\&= 0.22 + 0.08 \\&= \boxed{0.30}\end{aligned}$$

About 30% of Dalmatians have hearing impairment in at least one ear. We can check this using the complement:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - 0.70 = 0.30$$

At least one versus more than one

$P(X \geq 1) = 0.30$, while $P(X > 1) = P(X = 2) = 0.08$.

Example 9.8: Probability of Some Hearing Impairment

The Dalmatian probability model can be represented with two short vectors.

```
# Possible values of X
x <- c(0, 1, 2)

# Corresponding probabilities
p_x <- c(0.70, 0.22, 0.08)

# Check that this is a valid probability model
sum(p_x)
# 1

# Probability of at least one impaired ear
sum(p_x[x >= 1])
# 0.30
```

Connection to the mathematics

For a discrete model, the probability of an event is found by **adding the probabilities of the outcomes inside the event**.

Break Time



Continuous Probability Models

A continuous sample space contains an entire interval of possible values, so we cannot assign probability by listing individual points.

Density curve

A density curve:

- 1 lies on or above the horizontal axis;
- 2 has total area exactly equal to 1.

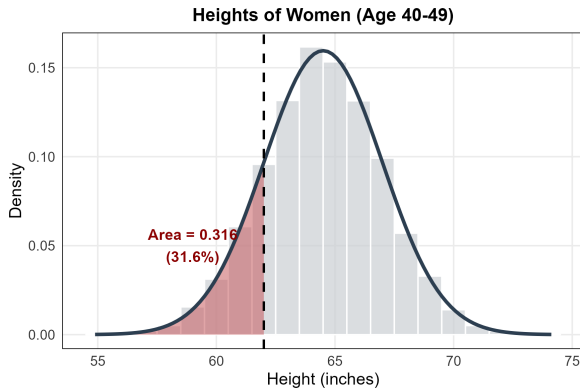
For a continuous random variable Y ,

$$P(a \leq Y \leq b) = \text{area under the density curve from } a \text{ to } b.$$

Important

Probability is represented by **area**, not by the height of the curve at one particular value.

Example 9.9–9.10: Patients' Heights



The histogram shows simulated patient heights. The smooth curve is a continuous probability model for this patient population.

- The total area under the curve is 1.
- The shaded area to the left of 62 inches represents the probability that a randomly selected patient is shorter than 62 inches.
- In this model, that probability is approximately 0.316.

Probability as Area

For the model of patient's heights shown previously

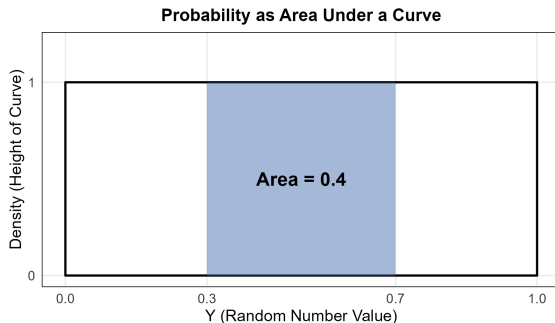
$$P(Y \leq 62) = 0.316$$

- The probability is an **area** not the height of the curve.
- The total area under the curve is 1.
- The shaded area represents 31.6% of the modeled population.

Example 9.11: Uniform Density

For a uniform random number between 0 and 1, the density has constant height 1. Therefore, probability is especially easy to compute:

$$\text{area} = \text{width} \times \text{height}.$$



$$\begin{aligned} P(0.3 \leq Y \leq 0.7) &= (0.7 - 0.3)(1) \\ &= \boxed{0.4} \end{aligned}$$

The interval occupies 40% of the full width of $[0, 1]$, so it also contains 40% of the total probability.

Adding Disjoint Continuous Intervals

For $Y \sim \text{Uniform}(0, 1)$,

$$\begin{aligned} P(Y \leq 0.5 \text{ or } Y > 0.8) &= P(Y \leq 0.5) + P(Y > 0.8) \\ &= 0.5 + 0.2 = \boxed{0.7} \end{aligned}$$

Now suppose $X \sim \text{Uniform}(2, 10)$. The total interval has width 8, so

$$\begin{aligned} P(X < 4 \text{ or } X \geq 7) &= P(X < 4) + P(X \geq 7) \\ &= \frac{4 - 2}{8} + \frac{10 - 7}{8} = \boxed{\frac{5}{8}} \end{aligned}$$

The intervals in each example are disjoint, so we add their areas.

The Probability of an Exact Value

For a continuous random variable,

$$P(Y = c) = 0$$

for every exact value c .

Why?

A single point has zero width, so the area above that point is zero.

Consequently,

$$P(Y \leq 0.8) = P(Y < 0.8).$$

Parameters vs. Statistics

Sample statistics

Computed from observed sample data:

$$\bar{x} \quad s$$

Distribution parameters

Describe the theoretical probability distribution:

$$\mu \quad \sigma$$

The mean of a random variable is also called its **expected value**.

Mean of a Random Variable

Let X have possible values $x_1, \dots, x_k$ with probabilities $p_1, \dots, p_k$.

Expected value

$$\mu = E(X) = \sum_{i=1}^k x_i p_i$$

- The expected value is a **probability-weighted average**.
- Outcomes with larger probabilities contribute more to μ .
- The mean does not need to be a possible outcome.

Long-run interpretation

If we repeat the experiment many times, the average observed value tends toward μ .

Variance and Standard Deviation of a Random Variable

The variance measures how far the possible values of X lie from the mean μ , accounting for their probabilities.

$$\sigma^2 = \sum_{i=1}^k (x_i - \mu)^2 p_i \quad \sigma = \sqrt{\sigma^2}$$

- Each squared distance $(x_i - \mu)^2$ is weighted by its probability.
- Variance is measured in squared units.
- Standard deviation is measured in the same units as X .

Observed Data and a Probability Model

The calculations have the same goal: describe the centre and spread of a variable.

	Observed sample	Probability model
Mean	$\bar{x} = \frac{1}{n} \sum_{j=1}^n x_j$	$\mu = \sum_{i=1}^k x_i p_i$
Standard deviation	$s = \sqrt{\frac{\sum_{j=1}^n (x_j - \bar{x})^2}{n - 1}}$	$\sigma = \sqrt{\sum_{i=1}^k (x_i - \mu)^2 p_i}$

The connection

$\bar{x}$ and s describe the values we observed. μ and σ describe the probability model. With many independent observations from that model, $\bar{x}$ and s tend to get close to μ and σ .

Example 9.13: Expected Hearing Impairment

For $X \in \{0, 1, 2\}$ with probabilities 0.70, 0.22, 0.08

$$\begin{aligned}\mu = E(X) &= \sum_{i=1}^k x_i p_i \\ &= (0)(0.70) + (1)(0.22) + (2)(0.08) \\ &= \boxed{0.38}\end{aligned}$$

Interpretation

The expected value need not be a possible individual outcome. A dog cannot have 0.38 impaired ears, but 0.38 is the long-run average across many dogs.

Example 9.13: Variance and Standard Deviation

$$\begin{aligned}\sigma^2 &= \sum_{i=1}^k (x_i - \mu)^2 p_i \\ &= (0 - 0.38)^2(0.70) + (1 - 0.38)^2(0.22) + (2 - 0.38)^2(0.08) \\ &= 0.10108 + 0.084568 + 0.209952 \\ &= \boxed{0.3956} \\ \sigma &= \sqrt{0.3956} = \boxed{0.62897} \approx 0.63\end{aligned}$$

Probability, Risk and Odds

In the life sciences, the same uncertain event may be communicated using **probability**, **risk**, or **odds**. These quantities are related, but they are not numerically identical. If an event A occurs with probability p , then:

Risk

$$\text{Risk}(A) = p$$

In health applications, risk usually refers to the probability of an undesirable outcome.

Odds

$$\text{Odds}(A) = \frac{p}{1 - p}$$

Odds compare the probability that the event occurs with the probability that it does *not* occur.

Example 9.14: Probability of DVT

Patients with severely restricted mobility are at risk of deep vein thrombosis (DVT), a blood clot in a deep vein. Fragmin may be given to reduce that risk. Among 1518 patients receiving Fragmin, 42 experienced DVT.

Probability, or risk

$$p = P(\text{DVT}) = \frac{42}{1518} \approx 0.0277 = 2.77\%$$

So about 2.77% of these patients experienced DVT. The number who did not was

$$1518 - 42 = 1476$$

Example 9.14: Odds of DVT

Odds compare the number who experience an event with the number who do not:

$$\text{Odds of DVT} = 42 : 1476$$

Dividing both parts by 42 gives

$$42 : 1476 = 1 : \frac{1476}{42} \approx \boxed{1 : 35}$$

How to read 1 : 35

For approximately every one patient who experienced DVT, 35 patients did not.

Probability versus odds

The probability was 0.0277; the numerical odds were $\frac{42}{1476} \approx 0.0285$. These are close because DVT was rare in this group.

Example 9.15: Sickle-Cell Inheritance

Sickle-cell trait means a person carries one sickle-cell gene but does not have sickle-cell disease. If both parents have the trait, each can pass either their usual gene or their sickle cell gene to a child.

Possible outcomes for each child

- 25%: inherits neither sickle-cell gene
- 50%: inherits one and has sickle-cell trait
- 25%: inherits both and has sickle-cell disease

So the probability, or **risk**, of sickle-cell disease for each child is

$$p = 0.25 = 25\%$$

Example 9.15: Odds of Sickle-Cell Disease

The risk of disease is $p = 0.25$. The probability that a child does *not* have the disease is $1 - p = 0.75$.

$$\text{Odds} = \frac{p}{1 - p} = \frac{0.25}{1 - 0.25} = \frac{0.25}{0.75} = \frac{1}{3} \approx 0.333$$

The odds are 1 : 3: one child with the disease for every three without it, in the long-run ratio.

Risk and odds are different

The risk is 25% while the numerical odds are about 0.333.

Converting Odds Back to Probability

Starting from

$$\text{odds} = \frac{p}{1 - p}$$

solve for p :

$$p = \frac{\text{odds}}{1 + \text{odds}}$$

For odds of $1/3$

$$\begin{aligned} p &= \frac{1/3}{1 + 1/3} \\ &= \frac{1/3}{4/3} \\ &= \frac{1}{4} = 0.25 \end{aligned}$$

Key Takeaways

- Probability describes uncertainty using numbers from 0 to 1.
- Long-run relative frequency provides the frequentist interpretation.
- A probability model consists of a sample space and probability assignments.
- Disjoint events can be added; complements are found by subtracting from 1.
- Discrete probabilities are sums; continuous probabilities are areas.
- Random variables have theoretical means and standard deviations.
- Risk is a probability while odds are $p/(1 - p)$.